AI4XR: AI in Extended Reality for 3D Scene Editing and Accessibility Design

Junlong Chen
Department of Engineering
University of Cambridge
Cambridge, Cambridgeshire, United Kingdom
jc2375@cam.ac.uk

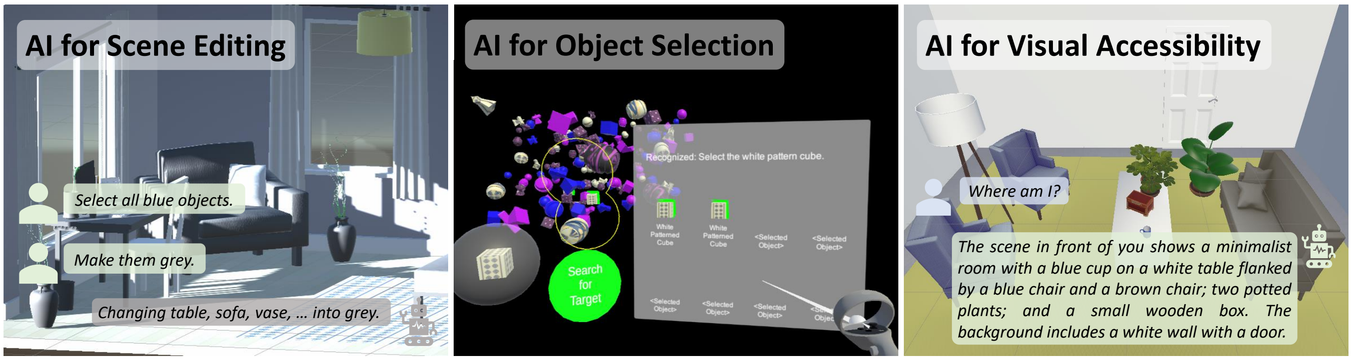

Figure 1: Examples of my current work showing how AI could be applied for scene editing [9] (left), object selection [8] (middle), and visual accessibility [7] (right) in immersive scenes. Left: Users edit the colors of multiple objects using the bulk modification strategy with AssistVR [9]. Middle: Users select multiple objects by referencing the color and shape of objects with AssistVR [8]. Right: Blind and low vision users obtain a natural language description of the virtual scene with EnVisionVR [7].

Abstract

Artificial Intelligence (AI) models have widespread applications for text, images, and videos. More recently, there has been increasing attention to applying AI models for 3D content and immersive scenes in extended reality (XR). My research explores the design tradeoffs of applying AI in XR from a user-centered perspective. Specifically, I draw upon applications such as immersive scene editing, occluded object selection, and visual accessibility features in virtual reality to discuss what constitutes a set of guidelines for the design of AI-assisted systems in XR environments. I explore the opportunities and challenges brought by the incorporation of AI compared with traditional interaction techniques in XR applications.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; **Accessibility systems and tools**; *Natural language interfaces.*

Keywords

Human-Computer Interaction, Virtual Reality, Extended Reality, Accessibility, Artificial Intelligence, Large Language Models

CHI EA '26, Barcelona, Spain

ACM ISBN 979-8-4007-2281-3/26/04
https://doi.org/10.1145/3772363.3799187

ACM Reference Format:
Junlong Chen. 2026. AI4XR: AI in Extended Reality for 3D Scene Editing and Accessibility Design. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26), April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3772363.3799187

1 Introduction

Artificial Intelligence (AI) models have a strong capability in understanding text, images, and videos, and are increasingly showing their potential in performing spatial reasoning and interpreting 3D scenes. My research aims to explore how such reasoning capabilities of AI models could be leveraged to support scene understanding and improve the interaction experience in extended reality (XR) environments. Through two application domains in AI-assisted 3D scene editing and AI-assisted visual accessibility design in virtual reality (VR), I demonstrate the strong potential of applying AI models in 3D environments and immersive scenes, and provide a foundation for future exploratory works.

Many works have studied how AI could be applied in spatial environments within the context of 3D scene editing and accessibility design tasks. These include for example, how AI could interpret and alter 3D scene representations such as mesh data or scene graphs, and how AI could provide descriptions of visual information to support accessibility design in spatial environments in the virtual and physical space. My research extends existing literature by focusing on how large language models could be integrated into existing scene editing and object manipulation workflows in immersive virtual reality scenes, and within this context, what are the design tradeoffs that emerge as a result of this new interaction paradigm. Additionally, I am also interested in how multimodal AI models such as vision-language models (VLMs) could support visual interpretation, and how these new technologies could be integrated with existing visual accessibility aids to provide a spectrum of high-level scene descriptions to low-level detailed object descriptions to provide accessibility features for blind and low vision users in virtual reality, and potentially in real-world environments.

2 Related Work

2.1 AI-Assisted 3D Scene Editing in Immersive Space

A number of recent works have explored the possibility of integrating AI and LLMs in extended reality environments [21, 43], as well as their implications on inclusion, engagement, and privacy [3]. These works can be classified into AI integration in different types of 3D representations with a stronger focus in AI models [34] and AI integration in immersive XR scenes with a focus in technical implementation and interaction design.

The availability of well-annotated datasets of 3D objects and 3D scenes such as Objaverse [14], 3D-Grand [46] and 3D-Front [17] allows AI models to be trained and systematically evaluated. Based on these datasets, researchers are able to train models and develop systems such as Holodeck [47] and SceneFormer [44] to use high-level representations to model and generate 3D scenes based on natural language input, or to generate or edit 3D objects using mesh representations (e.g., DreamMesh [45], Text2Mesh [37]). It is also possible to model 3D scenes using implicit representations like Neural Radiance Fields (NeRFs) [25, 52] and explicit representations like 3D Gaussian Splatting [16, 52]. Works like DreamFusion [41] and Magic3D [33] allow NeRF representations to be optimized to synthesize 3D models based on textual input, while SceneWeaver [22] and Gala3D [51] also demonstrate how it is possible to build coarse layouts based on textual prompts and use them to initialize and optimize 3D Gaussians to produce 3D scenes.

Many works also study how AI models can be integrated in immersive environments, and how does this affect user interaction behavior. A typical example is LLMR [13], a framework of multiple modules powered by specialized Generative Pre-trained Transformers (GPTs), which supports a wide range of use cases such as world creation and editing. Works such as MagicItem [29], DreamCodeVR [18] and SceneCraft [24] also leverage the strong capabilities of LLMs to generate code to render complex 3D scenes and define the behavior of virtual objects in the scene. Other examples such as LLMER [10] and VRCopilot [48] encode the 3D scene using parameters in structured JSON format and use GPT agents to interpret user natural language input and trigger scene edits. Based on this workflow, VRCopilot [48] explicitly studies how different levels of automation (manual, scaffolded, and automatic creation) affect user agency and creativity, while LLMER [10] demonstrates the effectiveness of the proposed system via different metrics. Other works have also studied how people prompt AI-assisted systems to create immersive VR scenes [1], as well as how other interaction modalities such as gestures could complement speech in LLM-based interaction in VR [23].

These works demonstrate an opportunity in applying state-of-the-art AI workflows for 3D content in XR spaces. Through user studies, we can identify interaction paradigms and failure cases to improve the interaction workflow by leveraging the rich interaction data and modalities in XR, which motivates my research.

2.2 AI for Visual Accessibility Design in Extended Reality

Accessibility design in extended reality has been a long-standing focus in existing literature [2, 12, 15, 38, 40]. Visual accessibility design is particularly challenging in XR due to the complexity of 3D scenes. Unlike 2D media where screen readers can be applied and a clear top-to-down and left-to-right order can be applied to read out elements on a planar user interface (UI), it is difficult to convey the large amount of information in virtual and physical 3D spaces. Numerous efforts have been made to address the challenge of providing visual accessibility features in XR environments. These efforts can be classified into two strategies, namely augmenting existing visual features, and converting visual features into other modalities such as audio cues, speech descriptions, and haptic feedback.

SeeingVR [50] and VRiAssist [36] are typical examples of the first strategy. These works use tools such as magnification and color/brightness/contrast adjustment to augment visual features of the original scene. The second strategy focuses on converting visual information to other forms. Regarding scene descriptions, WorldScribe [5] and WanderGuide [30] leverage AI models to provide visual descriptions of the physical world. Gonzalez et al. [19] provide a detailed summary of different use cases of AI-powered scene descriptions in the physical world. While not targeted specifically for accessibility, GazePointAR [32] provides a context-aware voice assistant in augmented reality by jointly analyzing eye gaze, pointing gestures, and natural language conversation. Works like SeeingVR [50], VRSight [27], and OmniScribe [6] also aim to provide scene/object-level descriptions or support for authoring descriptions in immersive environments. Many of these descriptions are powered by AI models which demonstrates the high potential of AI in XR visual accessibility applications. Visual information can also be converted to audio cues. VRBubble [26] used earcons, verbal notifications, and real-world sound effects to enhance the peripheral awareness of visually-impaired users. The effect of spatial audio cues versus speech feedback and grid positions was also compared by Lança et al. [31]. Last but not least, visual information can also be converted to haptic information, as demonstrated in Canetroller [49] and VIVR [28].

These works demonstrate how AI models can interpret 2D visual information and 3D spatial information for blind and low-vision users. Combined with various interaction modalities, AI advancements hold great potential in providing a more natural, intuitive, and embodied interaction experience for visual accessibility in extended reality.

3 Research Questions

While existing works demonstrate how AI models could be applied to various types of 3D representations, the application of AI models in immersive XR scenes is still at its early stage. Many works do exist, but most of them focus on a single technical pipeline to encode scene information using images or text and use proprietary LLMs accessed via APIs (e.g., the OpenAI API [39]) to process this information to generate parameters in structured JSON format or code to perform scene editing tasks. There is an opportunity to leverage the well-established different forms of scene representations in computer vision literature to combine with spatial interaction modalities such as gestures and raycasting to develop and test customized models in XR. Based on these new scene representations, we may overcome certain limitations of the existing workflow, where scenes are abstracted using text/images and processed by LLMs, before generating new text and recovered back into scenes. New interaction paradigms could provide guidelines for future designers who aim to integrate AI in XR environments to support scene understanding, editing, and accessibility features. Therefore, we formulate the following research questions (RQs):

- **RQ1:** Compared with traditional interaction modalities in 3D space, what are the advantages and disadvantages of intelligent speech-based interfaces in terms of user interaction with immersive 3D content?
- **RQ2:** What usability issues do current AI for XR workflows which use structured text to represent 3D scenes pose? Can interaction barriers be mitigated? If so, what are the mitigation strategies?
- **RQ3:** What categorizes an environment to deploy intelligent multimodal interaction systems and what are the emerging qualities arising through the deployment?

4 Methodology

My research adopts a user-centered design and evaluation approach. Throughout my past projects, my work involves designing a research prototype based on the proposed research question. This prototype system typically involves a baseline condition, and one to two experimental conditions. A number of participants are then invited to join a within-subjects user study, where each participant is exposed to all conditions. The order of conditions is counterbalanced across all participants to minimize learning effects. Throughout the study, performance data such as accuracy and task completion time are collected, and after completing each condition, participants are often invited to fill out usability [4], user experience [42], and task load [20] questionnaires. Quantitative data collected from interaction data and post-experience questionnaires are analyzed by conducting power analyses [11] and by running appropriate statistical tests. If significant differences exist between different conditions, these are reported and discussed. Qualitative data from questionnaire comments are often coded and analyzed using thematic analyses. Recurring themes from different participants are grouped together and reported in tables.

5 Progress to Date

My progress to date involves two publications at IEEE VR 2025 and ACM SUI 2025 in the first direction on AI for VR scene editing [8, 9] and one journal publication in IEEE TVCG in the second direction on AI for VR visual accessibility design [7]. The first two publications contribute towards **RQ1** and **RQ2**, while all three works contribute towards **RQ3**.

5.1 AI for Scene Editing in Virtual Reality

Within this scope, I have proposed AssistVR [9], a workflow which combines raycast selection with speech selection and editing. For the speech modality, a natural language processing module (Microsoft Azure Conversational Language Understanding) is used to identify the intent and extract key entities from the user speech input. If the intent is recognized as a query, the user natural language input is then sent to a large language model to provide a natural language response. Otherwise, based on the recognized intent and extracted key entities, Unity post-processing scripts are executed to perform downstream object selection and editing tasks in the Unity virtual scene.

Through an empirical user study ($N = 12$), participants were asked to edit the original indoor scene to match a target scene, where the target scene was specified using natural language instructions (such as instructing participants to make all blue objects purple) and additional visual instructions (images of the same target scene). The study identified typical interaction strategies, namely *incremental exploration* where participants inspected and made incremental edits to individual objects, and *bulk modification* where speech is used to bulk-select and bulk-edit a group of objects with shared properties (see Figure 1 left). This observation reflects a typical new interaction paradigm enabled by LLM-assisted systems in XR environments, where speech plays a pivotal role to support efficient multi-object interactions where shared properties are easily perceivable and easy to reference. Compared with traditional direct-manipulation systems, AI-assisted editing systems show the potential to perform rapid edits via simple speech commands.

In a second study, the AssistVR workflow was compared against a traditional minimap technique in occluded object selection tasks [8]. Here, the system was slightly modified where objects could be selected one-by-one using raycast, or selected together using speech by referencing the color and/or shape of the objects (see Figure 1 middle). The speech intent was classified as 'Select', 'Cancel All', or 'None', and key entities such as the object color and shape were extracted if they existed. Meanwhile, the minimap technique was adopted from DiscPIM [35] due to its strong performance in occluded object selection tasks among direct-manipulation techniques. Users use the left controller to direct the search cone in DiscPIM [35] to the region of interest to search for target objects. Objects within the search cone are projected on the disc, and the right controller can select objects from the disc. Cluttered objects on the disc can be further expanded along the disc circumference to be selected. A within-subjects user study ($N = 24$) was conducted to explore strengths and tradeoffs between speech-and-pointing and disocclusion minimap techniques. The study involved 3 levels of target object number (1/2/4 targets) and 3 levels of scene perplexity (low/medium/high). The scene perplexity is reflected by the number of object shapes and object colors which can easily be referenced verbally. For each combination of Technique × NumTargets × Perplexity, participants completed a search and repeat trial.

The results show that the minimap technique was significantly faster with single target selection ($p < .05$), while speech-and-pointing was significantly faster when there were 2 targets ($p < .05$) and 4 targets ($p < .001$). Participants reported a significantly lower overall load ($p < .05$) and moderately higher usability with speech-and-pointing, while a moderately better user experience was found with the minimap technique, with participants reporting DiscPIM [35] being 'fun', 'intuitive', 'more engaging', and providing 'a better sense of control and direct-manipulation'. Results from the study allow us to better understand the design tradeoffs of LLM-assisted workflows in XR environments. For example, the AI-powered speech-and-pointing interaction paradigm reduces overall load in multi-object selection tasks. It also requires careful design to provide high-quality speech recognition, allow users to reference objects easily via speech, and provide rich visual feedback to make the mapping between speech commands and subsequent actions explicit so as to improve user agency and overall experience.

5.2 AI for Visual Accessibility in Virtual Reality

Another direction of my current progress involves leveraging AI systems and multimodal interaction techniques to support visual accessibility design in virtual reality. This direction has yielded one journal publication. The proposed EnVisionVR system [7] is developed based on findings on accessibility barriers and adaptations for blind and low-vision participants in a formative study. Eight participants completed 34 sub-tasks in total, which were spread across two VR hardware tasks and five VR experiences. Participants highlighted the need for multimodal feedback to convey visual information, continuous descriptions to support scene understanding, binaural and 3D audio to convey spatial depth, and environmental audio cues to enhance immersion and provide feedback.

Based on findings from the formative study, we designed EnVisionVR, a scene interpretation tool for visual accessibility in virtual reality. The tool consisted of three core functions: (1) Scene Description, (2) Main Objects Indication, and (3) Object Localization. All three functions are activated by user voice commands.

Scene descriptions are generated before runtime by specifying a list of preset positions in the scene (referred to as 'anchor points'), and by uploading screenshots of the user camera view at eight different orientations along the horizontal plane to a vision-language model to generate natural language descriptions of the user's view. During runtime, the current user camera position and orientation is associated with the closest-matching anchor position and orientation to read out the pre-generated scene description. An example is shown in Figure 1 (right).

The Main Objects Indication Function assigns importance values to virtual objects in the scene before runtime. During runtime, a 'runtime importance value' is calculated for each object based on the original importance value and the distance between the object and the current user camera position. When the function is activated, the system reads out the object name and plays a short spatial tone for the three objects with the highest 'runtime importance value'. This allows users to locate important objects in the scene that are nearby. The 'runtime importance value' is then decayed for objects that have been read out to allow other objects to be introduced subsequently.

The Object Localization Function issues a beeping sound with a frequency inversely proportional to the distance between the controller and the object. When the controller is close enough to the object to interact with it, the controller vibrates to inform the user. When users hold up an object, a voice message is also played.

Results from an evaluation study with 12 blind and low vision participants reveal the strong potential of incorporating AI models like VLMs for accessibility design in XR environments. We formulate guidelines for future designs of such systems. For example, a hierarchy of high-level descriptions together with detailed object-level information could support users with different visual accessibility needs, anchor-based VLM descriptions for static scenes could provide descriptions with low latency, spatial reference frames should be consistent across different feedback modalities, and descriptions could be customized to further improve user experience.

6 Future Directions

Research questions proposed in Section 3 can be further investigated by exploring whether scene representations beyond structured text and images pose new challenges and opportunities from a user-centered design perspective. It is also worth expanding **RQ3** to critically analyze how the findings in a controlled VR environment might break down when these systems are deployed in the noisy and unpredictable physical world. There is also an opportunity to study the benefits of embodied interaction in AI for XR workflows, and how interaction modalities unique to XR provide support for 3D AI workflows, which would otherwise not be possible for workflows on 2D interfaces. It would be interesting to study the above topics from an application-oriented perspective to draw upon concrete design problems in accessibility design, interior design, urban planning, gaming, and and film production to explore the unique opportunities enabled by the integration of AI in XR environments.

7 Conclusion

My research demonstrates the potential of incorporating AI in extended reality by developing prototype systems for scene editing, object selection, and visual accessibility in virtual reality and evaluating them via user studies. This series of works contribute towards the research questions proposed in Section 3. They demonstrate that traditional speech interaction, when augmented with scene contextual information and sent to an AI agent to process, can support efficient multi-object selection and editing. When incorporated with other modalities such as raycast, interaction paradigms like speech-and-pointing can further enhance system usability. This motivates my future work to explore more integrations of interaction modalities with AI within XR environments, and how such integrations could support a wide variety of use cases. AI for accessibility in VR environments is a starting point, and I will extend this to AR and physical environments as well as other areas of application with more specific use cases and target user groups.

Acknowledgments

I would like to express my sincere gratitude to my supervisor, advisor, collaborators, lab mates, and family for their strong support. This work is supported by Cambridge Trust and the China Scholarship Council.

References

[1] Setareh Aghel Manesh, Tianyi Zhang, Yuki Onishi, Kotaro Hara, Scott Bateman, Jiannan Li, and Anthony Tang. 2024. How People Prompt Generative AI to Create Interactive VR Scenes. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. 2319–2340.

[2] Craig Anderton, Chris Creed, Sayan Sarcar, and Arthur Theil. 2025. Asleep at the Virtual Wheel: The Increasing Inaccessibility of Virtual Reality Applications. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–20.

[3] Efe Bozkir, Süleyman Özdel, Ka Hei Carrie Lau, Mengdi Wang, Hong Gao, and Enkelejda Kasneci. 2024. Embedding Large Language Models into Extended Reality: Opportunities and Challenges for Inclusion, Engagement, and Privacy. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces*. 1–7.

[4] John Brooke et al. 1996. SUS-A quick and dirty usability scale. *Usability evaluation in industry* 189, 194 (1996), 4–7.

[5] Ruei-Che Chang, Yuxuan Liu, and Anhong Guo. 2024. WorldScribe: Towards Context-Aware Live Visual Descriptions. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–18.

[6] Ruei-Che Chang, Chao-Hsien Ting, Chia-Sheng Hung, Wan-Chen Lee, Liang-Jin Chen, Yu-Tzu Chao, Bing-Yu Chen, and Anhong Guo. 2022. OmniScribe: Authoring Immersive Audio Descriptions for 360° Videos. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–14.

[7] Junlong Chen, Rosella P. Galindo Esparza, Vanja Garaj, Per Ola Kristensson, and John Dudley. 2025. EnVisionVR: A Scene Interpretation Tool for Visual Accessibility in Virtual Reality . *IEEE Transactions on Visualization & Computer Graphics* 01 (Oct. 2025), 1–14. doi:10.1109/TVCG.2025.3617147

[8] Junlong Chen, Jens Grubert, and Per Ola Kristensson. 2025. A Comparative Study of Speech-and-Pointing and Disocclusion Mini-Map Techniques for Object Selection in Virtual Reality. In *Proceedings of the 2025 ACM Symposium on Spatial User Interaction*. 1–12.

[9] Junlong Chen, Jens Grubert, and Per Ola Kristensson. 2025. Analyzing Multimodal Interaction Strategies for LLM-Assisted Manipulation of 3D Scenes. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 206–216.

[10] Jiangong Chen, Xiaoyi Wu, Tian Lan, and Bin Li. 2025. LLMER: Crafting Interactive Extended Reality Worlds with JSON Data Generated by Large Language Models. *IEEE Transactions on Visualization and Computer Graphics* (2025).

[11] Jacob Cohen. 2013. *Statistical Power Analysis for the Behavioral Sciences*. routledge.

[12] Chris Creed, Maadh Al-Kalbani, Arthur Theil, Sayan Sarcar, and Ian Williams. 2024. Inclusive AR/VR: Accessibility Barriers for Immersive Technologies. *Universal Access in the Information Society* 23, 1 (2024), 59–73.

[13] Fernanda De La Torre, Cathy Mengying Fang, Han Huang, Andrzej Banburski-Fahey, Judith Amores Fernandez, and Jaron Lanier. 2024. LLMR: Real-time Prompting of Interactive Worlds using Large Language Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–22.

[14] Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli VanderBilt, Ludwig Schmidt, Kiana Ehsani, Aniruddha Kembhavi, and Ali Farhadi. 2023. Objaverse: A Universe of Annotated 3D Objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 13142–13153.

[15] John Dudley, Lulu Yin, Vanja Garaj, and Per Ola Kristensson. 2023. Inclusive Immersion: A review of efforts to improve accessibility in virtual reality, augmented reality and the metaverse. *Virtual Reality* 27, 4 (2023), 2989–3020.

[16] Ben Fei, Jingyi Xu, Rui Zhang, Qingyuan Zhou, Weidong Yang, and Ying He. 2024. 3D Gaussian Splatting as a New Era: A Survey. *IEEE Transactions on Visualization and Computer Graphics* (2024).

[17] Huan Fu, Bowen Cai, Lin Gao, Ling-Xiao Zhang, Jiaming Wang, Cao Li, Qixun Zeng, Chengyue Sun, Rongfei Jia, Binqiang Zhao, et al. 2021. 3D-FRONT: 3D Furnished Rooms with layOuts and semaNTics. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 10933–10942.

[18] Daniele Giunchi, Nels Numan, Elia Gatti, and Anthony Steed. 2024. DreamCodeVR: Towards Democratizing Behavior Design in Virtual Reality with Speech-Driven Programming. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 579–589.

[19] Ricardo E Gonzalez Penuela, Jazmin Collins, Cynthia Bennett, and Shiri Azenkot. 2024. Investigating Use Cases of AI-Powered Scene Description Applications for Blind and Low Vision People. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–21.

[20] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, 139–183.

[21] Teresa Hirzle, Florian Müller, Fiona Draxler, Martin Schmitz, Pascal Knierim, and Kasper Hornbæk. 2023. When XR and AI Meet - A Scoping Review on Extended Reality and Artificial Intelligence. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–45.

[22] Xiaolu Hou, Mingcheng Li, Jiawei Chen, Dingkang Yang, Ziyun Qian, and Lihua Zhang. 2024. SceneWeaver: Text-Driven Scene Generation with Geometry-aware Gaussian Splatting. In *The 16th Asian Conference on Machine Learning (Conference Track)*.

[23] Xiyun Hu, Dizhi Ma, Fengming He, Zhengzhe Zhu, Shao-Kang Hsia, Chenfei Zhu, Ziyi Liu, and Karthik Ramani. 2025. GesPrompt: Leveraging Co-Speech Gestures to Augment LLM-Based Interaction in Virtual Reality. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. 59–80.

[24] Ziniu Hu, Ahmet Iscen, Aashi Jain, Thomas Kipf, Yisong Yue, David A Ross, Cordelia Schmid, and Alireza Fathi. 2024. Scenecraft: An LLM agent for synthesizing 3D scenes as blender code. In *Forty-first International Conference on Machine Learning*.

[25] Arthur Hubert, Gamal Elghazaly, and Raphaël Frank. 2025. Editing Implicit and Explicit Representations of Radiance Fields: A Survey. *Machine Vision and Applications* 36, 6 (2025), 119.

[26] Tiger F Ji, Brianna Cochran, and Yuhang Zhao. 2022. VRBubble: Enhancing Peripheral Awareness of Avatars for People with Visual Impairments in Social Virtual Reality. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–17.

[27] Daniel Killough, Justin Feng, Zheng Xue Ching, Daniel Wang, Rithvik Dyava, Yapeng Tian, and Yuhang Zhao. 2025. VRSight: An AI-Driven Scene Description System to Improve Virtual Reality Accessibility for Blind People. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. 1–17.

[28] Jinmo Kim. 2020. VIVR: Presence of Immersive Interaction for Visual Impairment Virtual Reality. *IEEE Access* 8 (2020), 196151–196159.

[29] Ryutaro Kurai, Takefumi Hiraki, Yuichi Hiroi, Yutaro Hirao, Monica Perusquía-Hernández, Hideaki Uchiyama, and Kiyoshi Kiyokawa. 2025. MagicItem: Dynamic Behavior Design of Virtual Objects With Large Language Models in a Commercial Metaverse Platform. *IEEE Access* (2025).

[30] Masaki Kuribayashi, Kohei Uehara, Allan Wang, Shigeo Morishima, and Chieko Asakawa. 2025. WanderGuide: Indoor Map-less Robotic Guide for Exploration by Blind People. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–21.

[31] Diogo Lança, Manuel Piçarra, Inês Gonçalves, Uran Oh, André Rodrigues, and João Guerreiro. 2024. Speed-of-Light VR for Blind People: Conveying the Location of Arm-Reach Targets. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–5.

[32] Jaewook Lee, Jun Wang, Elizabeth Brown, Liam Chu, Sebastian S Rodriguez, and Jon E Froehlich. 2024. GazePointAR: A Context-Aware Multimodal Voice Assistant for Pronoun Disambiguation in Wearable Augmented Reality. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–20.

[33] Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. 2023. Magic3D: High-Resolution Text-to-3D Content Creation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 300–309.

[34] Xianzheng Ma, Brandon Smart, Yash Bhalgat, Shuai Chen, Xinghui Li, Jian Ding, Jindong Gu, Dave Zhenyu Chen, Songyou Peng, Jia-Wang Bian, et al. 2024. When LLMs step into the 3D World: A Survey and Meta-Analysis of 3D Tasks via Multi-modal Large Language Models. *arXiv preprint arXiv:2405.10255* (2024).

[35] Mykola Maslych, Yahya Hmaiti, Ryan Ghamandi, Paige Leber, Ravi Kiran Kattoju, Jacob Belga, and Joseph J LaViola. 2023. Toward Intuitive Acquisition of Occluded VR Objects Through an Interactive Disocclusion Mini-map. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 460–470.

[36] Sina Masnadi, Brian Williamson, Andrés N Vargas González, and Joseph J LaViola. 2020. VRiAssist: An Eye-Tracked Virtual Reality Low Vision Assistance Tool. In *2020 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW)*. IEEE, 808–809.

[37] Oscar Michel, Roi Bar-On, Richard Liu, Sagie Benaim, and Rana Hanocka. 2022. Text2Mesh: Text-Driven Neural Stylization for Meshes. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 13492–13502.

[38] Martez Mott, Edward Cutrell, Mar Gonzalez Franco, Christian Holz, Eyal Ofek, Richard Stoakley, and Meredith Ringel Morris. 2019. Accessible by Design: An Opportunity for Virtual Reality. In *2019 IEEE international symposium on mixed and augmented reality adjunct (ISMAR-adjunct)*. IEEE, 451–454.

[39] OpenAI. 2026. API Reference - OpenAI API. https://platform.openai.com/docs/api-reference/introduction

[40] Achraf Othman, Khansa Chemnad, Aboul Ella Hassanien, Ahmed Tlili, Christina Yan Zhang, Dena Al-Thani, Fahriye Altınay, Hajer Chalghoumi, Hend S. Al-Khalifa, Maisa Obeid, et al. 2024. Accessible Metaverse: A Theoretical Framework for Accessibility and Inclusion in the Metaverse. *Multimodal technologies and interaction* 8, 3 (2024), 21.

[41] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. 2022. DreamFusion: Text-to-3D using 2D Diffusion. arXiv:2209.14988 [cs.CV] https://arxiv.org/abs/2209.14988

[42] Martin Schrepp, Andreas Hinderks, and Jörg Thomaschewski. 2017. Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S). *International Journal of Interactive Multimedia and Artificial Intelligence* 4 (01 2017), 103. doi:10.9781/ijimai.2017.09.001

[43] Yiliu Tang, Jason Situ, Andrea Yaoyun Cui, Mengke Wu, and Yun Huang. 2025. LLM Integration in Extended Reality: A Comprehensive Review of Current

Trends, Challenges, and Future Perspectives. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–24.

[44] Xinpeng Wang, Chandan Yeshwanth, and Matthias Nießner. 2021. SceneFormer: Indoor Scene Generation with Transformers. In *2021 International Conference on 3D Vision (3DV)*. IEEE, 106–115.

[45] Haibo Yang, Yang Chen, Yingwei Pan, Ting Yao, Zhineng Chen, Zuxuan Wu, Yu-Gang Jiang, and Tao Mei. 2024. DreamMesh: Jointly Manipulating and Texturing Triangle Meshes for Text-to-3D Generation. In *European Conference on Computer Vision*. Springer, 162–178.

[46] Jianing Yang, Xuweiyi Chen, Nikhil Madaan, Madhavan Iyengar, Shengyi Qian, David F Fouhey, and Joyce Chai. 2025. 3D-GRAND: A Million-Scale Dataset for 3D-LLMs with Better Grounding and Less Hallucination. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 29501–29512.

[47] Yue Yang, Fan-Yun Sun, Luca Weihs, Eli VanderBilt, Alvaro Herrasti, Winson Han, Jiajun Wu, Nick Haber, Ranjay Krishna, Lingjie Liu, et al. 2024. Holodeck: Language Guided Generation of 3D Embodied AI Environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16227–16237.

[48] Lei Zhang, Jin Pan, Jacob Gettig, Steve Oney, and Anhong Guo. 2024. VRCopilot: Authoring 3D Layouts with Generative AI Models in VR. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–13.

[49] Yuhang Zhao, Cynthia L Bennett, Hrvoje Benko, Edward Cutrell, Christian Holz, Meredith Ringel Morris, and Mike Sinclair. 2018. Enabling People with Visual Impairments to Navigate Virtual Reality with a Haptic and Auditory Cane Simulation. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–14.

[50] Yuhang Zhao, Edward Cutrell, Christian Holz, Meredith Ringel Morris, Eyal Ofek, and Andrew D Wilson. 2019. SeeingVR: A Set of Tools to Make Virtual Reality More Accessible to People with Low Vision. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–14.

[51] Xiaoyu Zhou, Xingjian Ran, Yajiao Xiong, Jinlin He, Zhiwei Lin, Yongtao Wang, Deqing Sun, and Ming-Hsuan Yang. 2024. GALA3D: Towards Text-to-3D Complex Scene Generation via Layout-guided Generative Gaussian Splatting. In *Proceedings of the 41st International Conference on Machine Learning* (Vienna, Austria) *(ICML'24)*. JMLR.org, Article 2570, 11 pages.

[52] Chen-Yang Zhu, Xin-Yao Liu, Kai Xu, and Ren-Jiao Yi. 2026. A survey on 3D editing based on NeRF and 3DGS. *Frontiers of Computer Science* 20, 4 (2026), 2004701.